What changes an asteroid’s path?

antocciphysics | TEACHER KEY | CPA

Learning target: Explain how gravity can change velocity without contact.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 20; 3 course days later.*
$a=5$ m/s² right; $v_f=10$ m/s right. Constant net force and mass throughout.

02. Explain a worked example (worked)
Result points up and right; an upward force does not instantly erase rightward motion.

03. Look and explain (model)
Velocity is tangent; force points toward the central body. Force changes velocity rather than simply pointing along motion.

04. Build the solution (practice)
4 m/s right and 1 m/s down; diagonal down-right.

05. Use the idea (reason)
Space near Earth has gravity. Astronauts and spacecraft fall together, so there is little supporting normal force.

06. Choose a method (mixed) *Review: Unit 1, Day 11; 12 course days later.*
7 m/s²; the mass doubles at the same net force. Keep the net force fixed, including resistance.

07. Exit check (exit)
A curved path/turn changes direction, therefore velocity.

08. STRETCH: Apply it another way (stretch)
Longer action produces a larger upward velocity change, assuming the same mass and constant force.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 20; 3 course days later.*
x interval [-33, -27] km; y interval [45, 55] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

02. Explain a worked example (worked)
Result points up and right; an upward force does not instantly erase rightward motion.

03. Represent and interpret (model)
Velocity is tangent; force points toward the central body. Force changes velocity rather than simply pointing along motion.

04. Solve and justify (practice)
v=(4,-2) m/s; speed √20=4.47 m/s, increased from 4. Constant downward a is not uniform circular motion.

05. Analyze the reasoning (reason)
Space near Earth has gravity. Astronauts and spacecraft fall together, so there is little supporting normal force.

06. Choose a method (mixed) *Review: Unit 1, Day 11; 12 course days later.*
Net force 0.35 N; new a≈0.3889 m/s². Extra resistance reduces net force and acceleration if driving force is unchanged.

07. Exit check (exit)
Direction changes; speed stays constant because perpendicular force does no work. Velocity is not constant.

08. STRETCH: Apply it another way (stretch)
Velocity (8, 6) m/s; speed 10 m/s. Original horizontal component remains.